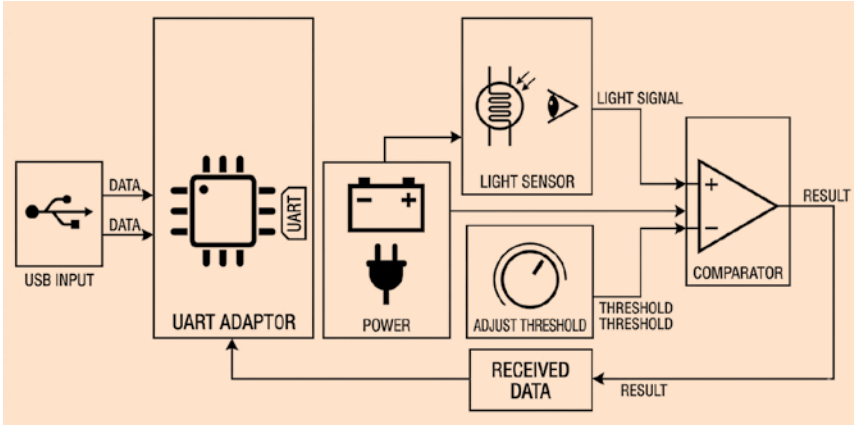


Fig. 2: Block diagram

reliable only at about 30ms per interval or slower, roughly 30-40 bps. The limit is the phone itself: the yellow phosphor layer that turns a white LED's blue light into white responds slowly, and operating-system latency adds more delay.

Fig. 1 shows the author's breadboard prototype, while Fig. 2 presents the project's block diagram.

What will you learn?

- How information can be sent wirelessly as visible light from a smartphone flashlight and received with a photodiode, using Morse code and OOK.
- How comparators and threshold-based signal conditioning turn an optical signal into clean digital pulses.
- Why timing analysis, pulse-width detection, and gap measurement matter when decoding real-world signals.
- How to connect hardware to Python through a USB-to-TTL converter and its serial control lines.
- How a discrete circuit and a software algorithm combine into a complete proof-of-concept optical link.

READY TO BUILD IT YOURSELF?

Head to **https://efy.co.in/gigh53** for the complete circuit diagram, component list, step-by-step build instructions, troubleshooting tips and demo videos. Everything you need to take this project from idea to working prototype is waiting for you there!

Abhay Verma is an electronics engineer at EFY and a passionate electronics hobbyist with a keen interest in practical circuit design and DIY projects.

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